



Ride-Sharing Analytics Platform

Review – 3

Section No : 04

TEAM NO : 20

2420090101 – N DhanuSree

2420090032 - Y Viditha

2420030678 – G GopiKrishna

Under the guidance of

Dr. NALLA SIRISHA

Associate Professor

Department of Computer Science and Engineering



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Table of Contents

- Abstract
- Introduction
- Objectives
- Literature Survey
- System architecture

Abstract

The rapid growth of ride-sharing platforms like Uber has made data-driven decision-making essential for improving service efficiency and customer experience in urban transportation. This project, "Ride-Sharing Analytics Platform," proposes an Uber-like system that stores trip information, calculates peak traffic hours, tracks driver performance, and predicts future ride demand. The platform is built using Python for data processing, PostgreSQL for structured storage, Apache Kafka for real-time streaming, Apache Airflow for ETL pipeline orchestration, Docker for containerized deployment, and Power BI for interactive dashboard visualization. By automating data collection and transformation through scalable ETL pipelines, the system delivers actionable insights on demand patterns, traffic trends, and driver performance, supporting better fleet management and pricing decisions. Through this project, practical skills are gained in ETL pipeline development, database management, data warehousing, and dashboard design — key competencies for real-world data engineering roles.

Introduction

- The emergence of ride-sharing platforms has significantly reshaped urban mobility, offering passengers a convenient, technology-driven alternative to traditional modes of transportation.
- These platforms generate massive volumes of data daily — including trip requests, real-time driver locations, fare details, and user behavior patterns — making efficient data analysis critical for service improvement.
- Despite this abundance of data, many platforms lack a structured analytical framework to convert raw data into actionable business insights.
- Common operational challenges include inconsistent driver availability during peak hours, inefficient route allocation, and unpredictable fare surges, all of which affect customer satisfaction and platform reliability.
- This project addresses these challenges by developing a Ride-Sharing Analytics Platform that processes trip data through automated pipelines to generate insights on demand, traffic, and driver performance.
- The system leverages modern data engineering tools to deliver a scalable, real-time analytics solution that supports informed, data-driven decision-making

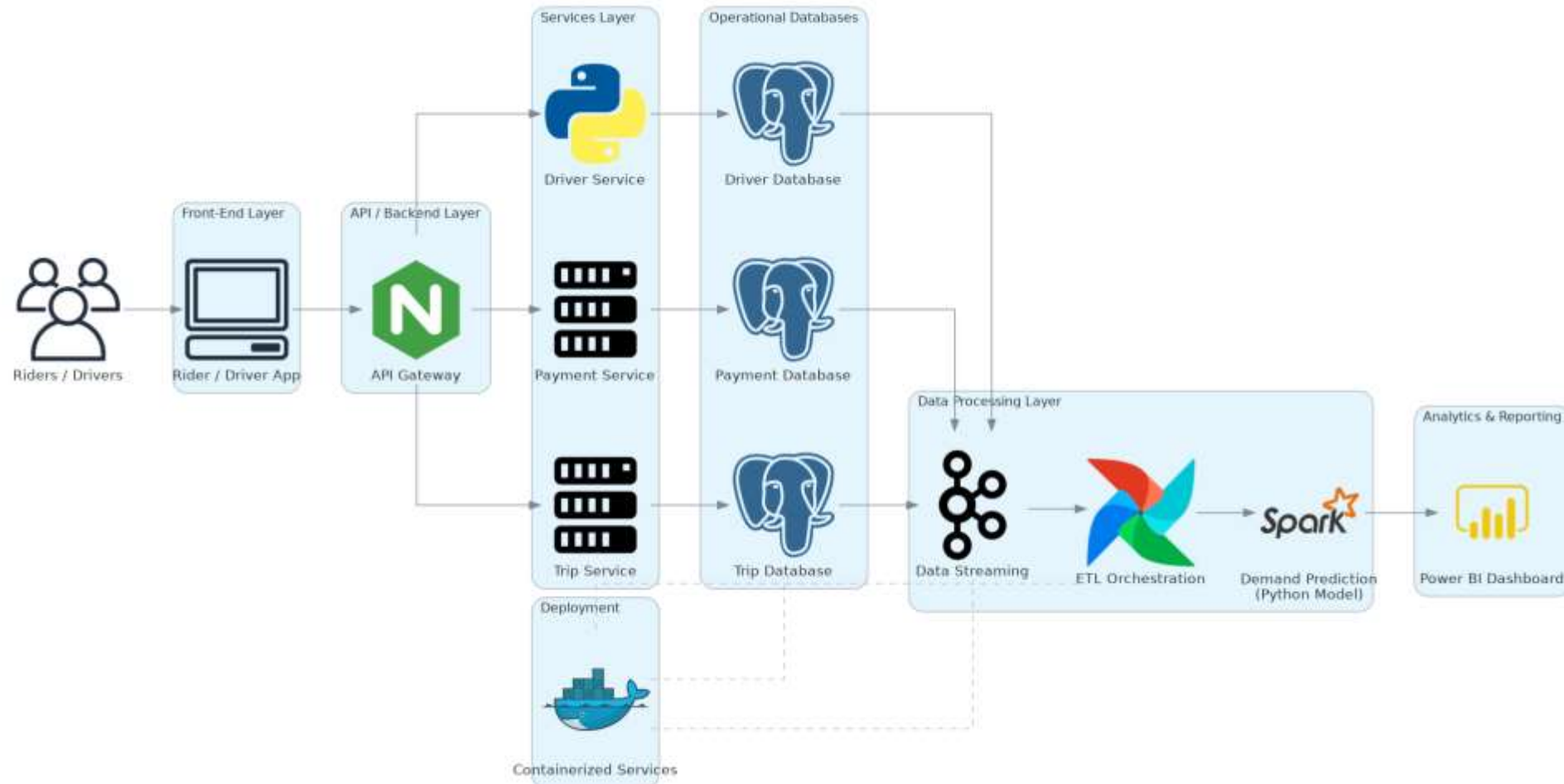
Objectives

- To design and develop a data analytics platform that stores comprehensive ride-sharing trip information (pickup/drop location, fare, duration)
- To calculate and visualize peak traffic hours across different zones for better fleet planning
- To build a module for tracking and evaluating individual driver performance
- To implement a predictive model that forecasts future ride demand using historical data
- To build automated ETL pipelines using Apache Airflow for reliable, scheduled data processing
- To enable real-time data ingestion using Apache Kafka for continuous trip event streaming
- To containerize the entire application using Docker for consistent, scalable deployment
- To present all insights through an interactive Power BI dashboard for data-driven decision-making

Literature Survey

S.No	Research paper	Author(s)	Link	Drawback
1.	Short-term prediction of demand for ride-hailing services: A deep learning approach (UberNet)	Long Chen et al.	https://arxiv.org/abs/2212.03956	Trained on NYC-specific data; may not generalize well to other cities with different traffic patterns
2.	Analysis and prediction of New York City taxi and Uber demands	(NYC study, published in scielo.org.mx)	scielo.org.mx/scielo.php?script=sci_arttext&pid=S1665-64232023000500886	Built around NYC's zip-code zones and weather data; not directly applicable to other cities
3.	Ride-hailing fares and demand interactions	(ScienceDirect, 2026)	https://sciencedirect.com/science/article/abs/pii/S0739885926000351	Focuses only on Uber and Lyft in New York City over a short 15-day window, so findings may not hold for other markets or longer time periods.
4.	Predicting rideshare demand (Public Policy Analytics, Ch. 8)	Ken Steif	https://urbanspatial.github.io/PublicPolicyAnalytics	Notes that spatio-temporal forecasting remains an open research problem, and reactive routing approaches often arrive after a demand spike has already ended.
5.	Modeling and prediction of ride-sharing utilization dynamics	(Hindawi, Journal of Advanced Transportation)	https://hindawi.com/journals/jat/2019/6125798	The prediction method is validated on a single dataset of NYC taxi trips from a specific month, limiting confidence in broader applicability.
6.	Real-time text analytics pipeline using open-source big data tools	Hassan Nazeer, Waheed Iqbal, Fawaz Bokhari, Faisal Bukhari, Shuja Ur Rehman Baig	https://arxiv.org/pdf/1712.04344	Evaluated only on Twitter sentiment data at a relatively small scale (under half a million tweets), so scalability to much larger, continuous ride-sharing data streams is not proven.

System Architecture



Ride-Sharing Analytics Platform Architecture

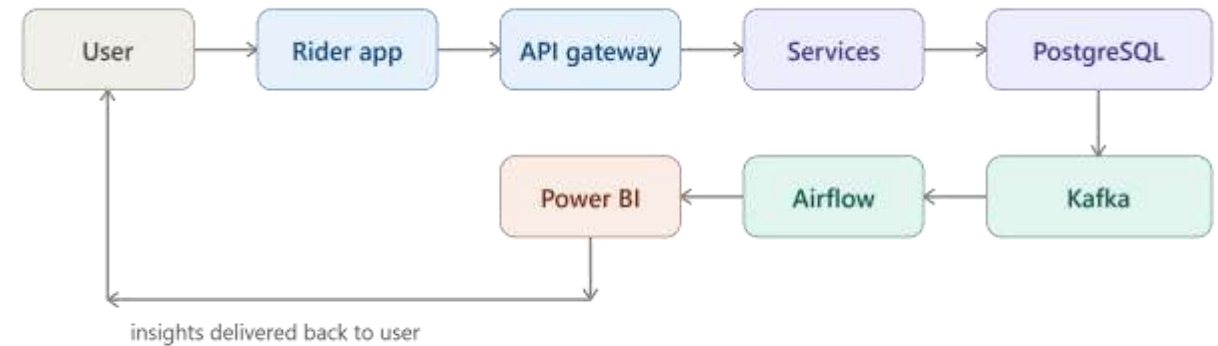
Methodology, System Design and Architecture

Methodology

- Collect trip, driver, and payment data from the app
- Stream real-time trip events using Kafka
- Clean and transform data via automated Airflow pipelines
- Analyze traffic, driver performance, and demand trends
- Display insights on Power BI dashboards

System Design

- Riders/drivers generate trip, driver, and payment events
- API Gateway routes requests to backend services
- Services store structured data in PostgreSQL
- Pipeline streams and transforms data in real time
- Analytics engine produces traffic and demand insights



Use of Modern Tools and Technologies

Technologies Used

- **Processing:** Python – data processing and demand prediction logic
- **Storage:** PostgreSQL – structured storage for trip, driver, and payment data
- **Streaming:** Apache Kafka – real-time streaming of trip events
- **Automation:** Apache Airflow – orchestration of automated ETL pipelines
- **Deployment:** Docker – containerized deployment of backend services
- **Visualization:** Power BI – interactive dashboards for insights
- These tools are widely used in real-world data engineering roles at companies like Uber, Netflix, and Amazon — reflecting industry-standard practices rather than academic-only methods.

Benefits

- Efficient, real-time data processing at scale.
- Better analysis of traffic and demand patterns.
- Supports accurate driver performance tracking.
- Helps improve fleet management and decision-making.

Implementation, Coding Quality and Functionality

Implementation

- Build ETL and streaming pipeline
- Process trip, driver, and payment data
- Analyze peak hours and driver performance
- Predict demand using historical data
- Display results on Power BI

Coding Quality

- Modular Python code (separate extract/transform/load functions)
- Clear variable/function naming
- Data validation before storage
- Independently testable components

Functionality → (insert a similar small flow image: Trip event → Kafka → Airflow → PostgreSQL → Analysis → Power BI, instead of the text sentence)

- Real-time trip/driver data collection
- Peak zone and hour detection
- Driver performance tracking
- Better fleet planning